

**Introduction:**

The UArizona Scholarship Application Management System (UASAMS) is a specialized software application program will be developed specifically for the University of Arizona, College of Engineering to streamline and automate the process of applying for and managing scholarships. This program will be used by the College of Engineering (ENGR) and non-profit organizations, corporations, private donors, and government agencies that offer scholarships to students.

This Concept of Operations (ConOps) document provides a comprehensive yet concise guide that helps project stakeholders understand the scholarship application software program's purpose, functionality, and operation. It serves as a valuable reference throughout the software development and implementation process. This ConOps document contains valuable information that can be used in the requirements development process for defining the software requirements for the system.

Objectives:

Scholarship application programs are valuable tools for organizations that want to efficiently manage scholarship programs, reduce administrative overhead, improve the applicant experience, and ensure fair and transparent scholarship selection processes. They help bridge the gap between scholarship providers and deserving students, making it easier for individuals to access financial support for their education.

Stakeholders:

The key stakeholders for this project include:

1. Scholarship providers
2. Applicants (students)
3. Applicant reviewers (Faculty, staff, and potentially donors)

4. College of Engineering administrators and staff
5. College of Engineering IT support staff
6. College of Engineering scholarship stewards
7. Application software development team
8. SFWE 301 professor

Note that other stakeholders may be identified as the product development activities proceed.

System Overview:

The UASAMS is intended to be a software application program designed to streamline and manage the process of applying for scholarships. Overall, the scholarship application software system aims to simplify the scholarship application process, reduce administrative overhead, and improve the overall efficiency and fairness of scholarship awarding. It is a valuable tool used by the College of Engineering, for organizations that offer scholarships, as well as for students seeking financial assistance for their education.

UASAMS should be accessible to users on several platforms including computers, tablets and mobile devices.

Operational Scenarios:

1. **Online Application Forms:** The system will provide a user-friendly interface for applicants to fill out scholarship application forms electronically. Applicants can input their personal information, academic achievements, extracurricular activities, and essay responses online.
2. **Document Upload:** Applicants can upload required documents such as transcripts, recommendation letters, essays, and other supporting materials directly to the system.
3. **Customization:** Organizations can customize the application forms and criteria to match their specific scholarship requirements. This allows for flexibility in tailoring scholarships to different purposes and target groups.
4. **Scholarship Generation:** Funding organizations and/or private parties can create new scholarships and specify the requirements for the created scholarship.
5. **Communication Tools:** The software will include communication tools for organizations to notify applicants of important updates, deadlines, and award decisions through email or notifications within the system.
6. **Review and Scoring:** Scholarship review committees can access the applications online and score them according to predefined criteria. This helps in the fair and efficient evaluation of applicants.
7. **Data Management:** The system collects and stores applicant data securely, making it easier for organizations to manage and analyze applicant information.
8. **Reporting and Analytics:** Scholarship administrators can generate reports and analyze application data to gain insights into applicant demographics, trends, and outcomes.

9. **Integration:** As part of a future release, the UASAMS should integrate with other software tools commonly used by the college, such as student information systems (SIS) or financial aid management systems.
10. **Payment Processing:** Once scholarship recipients are selected, the system will facilitate the disbursement of scholarship funds, ensuring that the financial aspect of the award is managed smoothly.
11. **Security and Compliance:** The system should prioritize data security and compliance with privacy regulations to protect applicants' and scholarship donors' sensitive information.
12. **User Support:** The application software system will offer user support and assistance to both applicants and administrators to ensure a smooth application and selection process.

System Requirements:

1. Application Management:

- 1.1. The system shall allow the College of Engineering to create and customize scholarship application forms that are available online.
- 1.2. The system shall prompt the applicant for the following information: name, student ID, NetID, degree major, degree minor (if applicable), academic achievements, financial information, and written essays.

2. User Interface:

- 2.1. The system shall provide a graphical user interface (GUI) that allows all users to have easy access to the system, including scholarship applicants, ENGR administrators and staff, reviewers, scholarship sponsors / donors, and scholarship stewards.

3. Document Submission:

- 3.1. Applicants shall be allowed to upload required documents such as transcripts, letters of recommendation, and essays directly into the system.
- 3.2. For creation of new scholarships, the system shall provide a scholarship donor or sponsor to enter the following information: scholarship name, scholarship description / requirements, recipient selection requirements, scholarship point of contact and contact information, scholarship amount, disbursement requirements, frequency of scholarship offering and other information and comments that they feel is relevant to the scholarship they are sponsoring.

4. Communication:

- 4.1. The system shall include communication features to send automated emails or notifications to applicants that inform them of application status updates, deadlines, and other relevant information.
- 4.2. The system shall include communications to current and prospective scholarship sponsors / donors. The communications include the following types of information: key

dates and deadlines for a new academic year of scholarship applications, notification of applicants for a given scholarship that they sponsor, and reports after scholarship awards have been made.

- 4.3. The system shall inform designated scholarship application reviewers that there are applicants to be reviewed for a specific scholarship.

5. **Application Review:**

- 5.1. The system shall allow scholarship administrators and reviewers access to evaluate applications online.
- 5.2. The system will pre-screen the applicants information to ensure that they meet the requirements of the scholarship. If the applicant does not meet requirements, they will be excluded from further consideration.
- 5.3. The system shall generate a list / report of applicants that meet all scholarship requirements for a given scholarship.
- 5.4. The system will allow reviewers to assign scores or ratings to applicants, make comments, and track the progress of each application.
- 5.5. The system will allow reviewers to ask the applicant for further clarifying information before they are selected or eliminated from further consideration.
- 5.6. The system shall allow a designated review panel to make the final selection of the applicant that the scholarship will be awarded to.

6. **Award Disbursement:**

- 6.1. Once scholarship recipients are selected, the system shall generate reports to facilitate the disbursement of funds.
- 6.2. As a future enhancement, the system shall integrate with financial aid systems and/or allow administrators to manually initiate payments to scholarship awardees.

7. **Reporting and Analytics:**

- 7.1. The system shall allow scholarship administrators to generate reports and analytics to gain insights into the application process, demographics of applicants, and the impact of scholarships awarded.

8. **Security and Compliance:**

- 8.1. The system shall have robust security features to protect sensitive applicant information and ensure compliance with data privacy regulations, (including FERPA, donor financial information, etc.).
- 8.2. The system shall require that all users have a valid username and password in order to access the UASAMS.
- 8.3. If a user does not have an authorized account on the UASAMS, the system shall prompt the user to create an account by entering a username and password which must be approved by a system administrator prior to having the account enabled.
- 8.4. The system shall prevent un-authorized users from gaining access to the system.

8.5. The system shall allow authorized users to only have access to portions of the system that they are granted permissions for.

9. **User Roles:** The system shall distinguish between different user roles, such as applicants, administrators, reviewers, and auditors, with varying levels of access and permissions. The table below defines the roles of different users (e.g., applicants, reviewers, administrators) and their responsibilities within the software system.

Type of User	Roles and Responsibilities
Student Applicants	Applicant for various scholarships available thru the UASAMS.
College of ENGR Scholarship Administrator	Notify potential applicants of key deadlines for scholarship applications, facilitate the overall scholarship selection process, notify recipients of scholarship awards, develop and oversee the mechanisms for disbursement of funds, maintain accurate records of scholarship awards, disbursements, and recipient information. Also ensures that the scholarship program complies with all relevant laws and regulations, including tax laws. Periodically evaluate the scholarship program's effectiveness and make improvements as needed. Solicit feedback from scholarship recipients and other stakeholders to identify areas for enhancement. Promotes diversity, equity, and inclusion in the scholarship program.
Applicant Reviewer	Participate in a fair and transparent selection process to evaluate applicants. This may involve forming a selection committee or using specific criteria to assess applicants' qualifications and potential.
Scholarship Provider	Scholarship organization's designee that plays a crucial role in supporting the educational pursuits of students by offering financial assistance to help them achieve their academic and career goals. The provider will define eligibility requirements, define the required application requirements for a given scholarship, optionally be involved in the scholarship selection process.
Scholarship Fund Steward	Responsibly manage and invest the funds to ensure the long-term sustainability of the program. Generate annual reports on the health and value of all donations / endowments. Ensure that funds are used for their intended purpose and provide periodic reports to stakeholders or donors, if applicable.
Authorized ENGR Staff	Support the Scholarship administrator in any delegated and authorized tasking.
ENGR IT Support Staff	Maintain the AUSAMS software product and make required enhancements or improvements to the tool as the program evolves from year to year. Also create and authorize new

	accounts for new college faculty, staff, and administrators, and delete accounts for those that are no longer involved in the College of Engineering scholarship application process.
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10. **Integration:** During the initial release of the UASAMS, integration with external databases, student information systems, or financial systems to streamline data exchange is not required. However, in future releases of the system, these features may be added.
11. **Scalability:** The system shall be scalable to accommodate an increasing number of applicants and scholarships as the program grows. This may include extended features to enhance user registration, application submission, application review, notifications, reporting, and data security.
12. **Training and Documentation:** The UASAMS shall provide user support and training resources to help administrators and applicants navigate the system effectively.
13. **Interfaces:** In the future, the system should be able to integrate with other software programs and databases that are used by the University of Arizona to include integration with financial aid databases, student records, applicant databases, or external communication channels.

For the Semester Project – Fall 2026:

- There will be teams of 3-4 students each.
 - All teams will develop their own complete Software Requirements Specification
 - All teams will be responsible for preparing for and conducting their own Acceptance Test Review
- Because of the size of the overall project, we will break implementation and test of the product into 4 primary subsystems:
 - **Front End** including:
 - Graphical User Interface for Applicants
 - GUI for Donors
 - GUI for ENGR Administrators
 - GUI for other ENGR staff
 - **Back End** including:
 - Data management of the Scholarship Applicants
 - Data management of the possible scholarships
 - Search engine that allows the user to either search for a given scholarship or a given applicant
 - **Matching Engine** including:
 - Algorithm to match applicants to scholarship requirements
 - Review and approval of matched scholarships
 - **Reports Engine** including:
 - Scholarship report(s)
 - Applicant report(s)
 - Candidate matching and review report
 - Final scholarship award disbursement report